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MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

IT-504D

OPERATIONS RESEARCH

Time Allotted: 3 Hours | Full Marks: 70

The questions are of equal value.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.
All symbols are of usual significance.

GROUP A
(Multiple Choice Type Questions)

1. Answer any *ten* questions. 10×1 = 10

(i) The following is not related to LPP

(A) Simplex Method (B) Transportation problem
(C) Hungarian Method (D) CPM

(ii) In an Assignment problem the minimum number of lines covering all zeros in the reduced cost matrix of order n can be

(A) at least n (B) $n + 1$
(C) $n - 1$ (D) at most n

(iii) Graphical Method is applicable for solving LPP

(A) of any type (B) with only two decision variables
(C) with only two constraints (D) none of these

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(iv) Following is a method to find final optimal solution of Transportation problem

(A) Hungarian Method (B) Matrix Minimum Method
(C) Stepping Stone Method (D) VAM

(v) If there are m source and n destination in a Transportation problem then number of constraint equations would be

(A) $m - n$ (B) mn
(C) $m - n + 1$ (D) $m + n - 1$

(vi) If in the Simplex Method, the basis column of the final table contains an artificial variable then the problem has _____ solution.

(A) unbounded (B) degenerate
(C) infeasible (D) multiple

(vii) CPM is

(A) probabilistic (B) deterministic
(C) event oriented (D) activity oriented

(viii) In a networking problem, the activity for which total float is zero is called

(A) independent activity (B) deterministic activity
(C) critical activity (D) none of these

(ix) In a game theory problem saddle point occurs when

(A) max(row min) = min(column max)
(B) min(row min) = min(column max)
(C) max(row min) = max(column max)
(D) none of these

(x) In a queue, if $(\lambda / \mu) > 1$ then the system is

(A) transient (B) system is in steady state
(C) system is in explosive state (D) none of these

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(xi) In PERT analysis, the variance of a job having optimistic time 5, pessimistic time 17 and most likely time 8, is

(A) 3 (B) 4
(C) 7 (D) 5

(xii) In a inventory model without shortage, if C_1 is the holding cost per unit per time, C_3 is the setup cost per order and R is the demand rate then EOQ is

(A) $(2C_3R/C_1)^{1/2}$ (B) $(2C_1R/C_3)^{1/2}$
(C) $(2C_1C_3R)^{1/2}$ (D) none of these

GROUP B
(Short Answer Type Questions)

Answer any *three* questions. 3×5 = 15

2. The chemistry department of a college requires 10, 12, 7 units of three chemicals X, Y, Z respectively. The chemicals are available in two types of boxes: Box A and Box B. Box A contains 3, 2, 1 units of X, Y, Z respectively and costs Rs. 300. Box B contains 1, 2, 3 units of X, Y, Z respectively and costs Rs.200. Formulate the LPP. Graphically show the feasible solution region and possible solution points.

3. Define the following :
(a) Zero Sum Game (b) Artificial Variable
(c) Holding Cost (d) Saddle Point
(e) Basic Variable

4. What is EOQ? Derive an EOQ model with uniform rate of demand, infinite production rate and having no shortage.

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5. Solve the following game graphically.

$$\text{Player A}\begin{pmatrix} 3 & -3 & 4 \\ -1 & 1 & -3 \end{pmatrix} \quad \text{Player B}$$

6. A company has three production facilities S1, S2, S3 with production capacity of 7, 9 and 18 units per week respectively which are to be shipped to four warehouse D1, D2, D3 and D4 with requirement of 5, 8, 7 and 14 units per week respectively. Cost matrix is given. Find the initial basic feasible solution using any method.

	D1	D2	D3	D4	Supply
S1	19	30	50	10	7
S2	70	30	40	60	9
S3	40	8	70	20	18
Demand	5	8	7	14	

GROUP C
(Long Answer Type Questions)

Answer any *three* questions. 3×15 = 45

7. Consider the following project activities with time estimates given in weeks.

Activity	Optimistic	Most likely	Pessimistic	Immediate Predecessor
A	1	3	7	-
B	2	6	14	A
C	3	3	3	A
D	4	10	22	B, C
E	3	7	15	B
F	2	5	14	D, E
G	4	4	4	D

(a) Draw the project network. 4
(b) Find the expected duration and variance of each activity. 2
(c) Calculate the early and late occurrence of each event and the expected project length. 4+2

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(d) What project duration will have 95% confidence of completion? 3

[Given that, $P(Z \le 1.64) = 0.95$]

8. (a) A department of a company has five employees with five jobs to be performed. The time (in hours) that each man takes to perform each job is given in effectiveness matrix. How should the jobs be allocated, one per employee, so as to minimize the total man-hour? 8

Employees

Jobs	I	II	III	IV	V
A	10	5	13	15	16
B	3	9	18	13	6
C	10	7	2	2	2
D	7	11	9	7	12
E	7	9	10	4	12

(b) Use Simplex method to solve the following LPP. 7

Maximize $Z = 5x_1 + 3x_2$

Subject to $3x_1 + 5x_2 \le 15$
$5x_1 + 2x_2 \le 10$
$x_1, x_2 \ge 0$

9. (a) Solve the following LPP using Big M method. 10

Minimize $Z = 4x_1 + 3x_2$

Subject to $x_1 + 2x_2 \ge 8$
$3x_1 + 2x_2 \ge 12$
$x_1, x_2 \ge 0$

(b) Find out the dual of the following problem. 5

Maximize $Z = 2x_1 + 3x_2 + 4x_3$

Subject to $3x_1 + x_2 + x_3 \le 2$
$-4x_1 + 3x_3 \ge 4$
$x_1 - 5x_2 + x_3 = 5$
$x_1, x_2 \ge 0$, x_3 is unrestricted in sign

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10. Three factories transport goods to five retail shops. The capacity of the factories and requirement of the shops are given in the table below. Find the final optimal solution in terms of amount to be transported from each factory to each shop so as to minimize the total transportation cost. Follow any methods of your choice. 15

	Shop1	Shop2	Shop3	Shop4	Shop5	Capacity
Factory1	1	9	13	36	51	50
Factory2	24	12	16	20	1	100
Factory3	14	33	1	23	26	150
Requirement	100	70	50	40	40	

11. (a) In a petrol pump customers arrive according to Poisson process with an average time of 5 min between arrivals. The service time is exponentially distributed with mean time = 2 min. 4×2

(i) What should be the average queue length?
(ii) What should be the average number of customers in the pump?
(iii) What is the average time spent by a car in the pump?
(iv) What is the average waiting time before receiving petrol?

(b) An item is used at a uniform rate of 50000 units per year. No shortage is allowed and delivery is at an infinite rate. Ordering, receiving and handling cost is Rs.13 per order while inspection cost is Rs.12 per order. Interest costs Rs 0.056 and deterioration and obsolescence cost Rs. 0.004 respectively per year for each item held in inventory plus Rs. 0.02 per year per unit based on the maximum number of units in inventory. Calculate EOQ and optimum number of orders per year. 7

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